HouseOpt-AI: A Coupled Predictive-Prescriptive Framework for Smart Hotel Housekeeping via Gradient-Boosted Demand Forecasting, Discrete Constraint Scheduling, and Monte Carlo Techno-Economic Modeling

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Abstract

Hotel housekeeping operations represent one of the most labor-intensive and economically volatile cost centers in the hospitality industry, directly accounting for 40% to 50% of property operating expenses. Existing housekeeping management protocols rely overwhelmingly on static, rule-based heuristics such as round-robin room dispatching and uncoordinated checkout servicing. These approaches suffer from three foundational structural flaws: inability to optimize sequential and spatial inter-floor transit latency, vulnerability to temporal lookahead leakage during turnover demand forecasting, and an absence of stochastic financial risk modeling under dynamic occupancy fluctuations. To resolve these challenges, we introduce HouseOpt-AI, an integrated predictive-prescriptive operational framework for smart hotel housekeeping. We evaluate our pipeline on 119,390 empirical reservations from the benchmark Hotel Booking Demand dataset under a strict chronological out-of-time partition (Train: 2015–2016, 521 days; Test: 2017, 257 days) calibrated to International Sanitary Supply Association (ISSA 540) cleaning standards and empirical hospitality wage scales. Our predictive engine, leveraging 32 engineered lag, rolling, and forward booking features, yields an out-of-time R^2 of 0.9269 (Ridge) and 0.8859 (LightGBM) with a Mean Absolute Error of 17.65 rooms/day, substantially outperforming classical 7-day moving-average benchmarks ($R^2 = 0.7823$, MAE $= 24.53$). Prescriptively, our discrete constraint optimization engine, formulated in Google OR-Tools CP-SAT over a SimPy discrete-event floor simulation of 120 rooms across 4 floors, reduces shift overtime by 34.03% ($p = 2.79 \times 10^{-60}$), accelerates floor makespan by 6.36% ($p = 2.90 \times 10^{-29}$), and diminishes room check-in Service Level Agreement (SLA) breaches from 3.89% to 0.89% (a 77.15% reduction, $p = 2.38 \times 10^{-7}$) across 30 simulated operational shifts. Finally, a 10,000-iteration Monte Carlo stochastic financial simulation demonstrates an average Cost Per Occupied Room (CPOR) reduction of INR 4.50 [95% CI: 0.13, 10.23], generating INR 157,450 in net annual labor savings for a standard 120-room property and achieving an amortized capital payback horizon of 16.4 months. Our findings demonstrate that mathematically coupling leakage-free machine learning demand forecasting with discrete constraint satisfaction programming delivers provable operational efficiency and robust financial feasibility in digital hospitality operations.

Keywords: Hotel Housekeeping Operations, Demand Workload Forecasting, Constraint Satisfaction Programming, Google OR-Tools CP-SAT, Discrete-Event Simulation, Monte Carlo

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Techno-Economics, Service Level Agreement (SLA).

1 INTRODUCTION

In global hospitality management, housekeeping operations represent the single largest ongoing operational expenditure, typically accounting for 40% to 50% of direct property operational budgets and exceeding hundreds of billions of dollars annually in aggregate labor allocations [1, 2]. Despite the critical economic and customer-satisfaction significance of room readiness, the physical coordination of housekeeping crews remains dominated by static, uncoordinated heuristics [3]. Operations managers routinely allocate tasks using clipboard manifests, fixed room-attendant sections, or greedy First-Come-First-Served (FCFS) dispatching rules as guests physically surrender room keys at the front desk [4].

Recent empirical analyses in service operations and hospitality technology have exposed three major structural flaws inherent in these static paradigms:

1. Inability to optimize sequential and spatial transit costs: Greedy and round-robin assignment mechanisms treat each vacated room as an isolated assignment event. They fail to account for the spatial topology of hotel buildings, frequently dispatching housekeepers across disparate floors. Consequently, room attendants spend up to 45 minutes per shift waiting for service elevators and traversing hallways, which amplifies physical worker fatigue, incurs unnecessary overtime penalties, and lengthens the overall floor cleaning makespan [2, 3].

2. Temporal demand volatility and future lookahead leakage: Daily housekeeping workload is driven by room turnover—the simultaneous departure of exiting guests and arrival of new guests. Classical historical moving averages fail to capture weekly seasonality spikes (e.g., weekend checkout waves) and length-of-stay dynamics. Furthermore, published forecasting studies frequently evaluate tabular machine learning models using random k-fold cross-validation, which inadvertently leaks future seasonal demand into historical training partitions, yielding deceptively high offline metrics ($R^2 > 0.95$) that fail when deployed on streaming operational rosters [5, 6].

3. Absence of stochastic financial feasibility modeling: Prior literature predominantly reports deterministic point estimates of cost savings (e.g., assuming a static 70% occupancy and fixed wage). Such estimates fail to convince hotel asset owners who face stochastic macroeconomic shocks, fluctuating occupancy rates, and inflationary labor pressures. Without rigorous confidence intervals derived through stochastic uncertainty modeling, the return on investment (ROI) of smart housekeeping retrofits remains mathematically unverified [10, 11].

To bridge this operational divide, this paper introduces HouseOpt-AI, an integrated, sequence-aware, predictive-prescriptive framework designed for scalable, real-time hotel housekeeping optimization. HouseOpt-AI couples tabular gradient-boosted demand forecasting, discrete-event floor operations modeling, constraint satisfaction programming, and stochastic techno-economic uncertainty analysis into a unified decision support pipeline.

The primary contributions of this paper are summarized as follows:

- Leakage-Free Chronological Workload Forecasting: We establish a strict chronological train/test partitioning protocol on 119,390 empirical hotel reservations, engineering 32 lag, rolling, and forward booking features. We demonstrate that Ridge regression ($R^2 = 0.9269$, MAE $= 17.65$) and LightGBM ($R^2 = 0.8859$, MAE $= 18.39$) significantly outperform classical moving-average baselines ($R^2 = 0.7823$, MAE $= 24.53$) without temporal data leakage.
- Spatial and Priority-Constrained Discrete Floor Optimization: We formulate hotel housekeeping task assignment as a discrete Constraint Satisfaction Problem (CSP) solved via Google OR-Tools CP-SAT over a SimPy discrete-event floor simulation. By enforcing non-overlapping intervals, shift overtime penalties, check-in SLA deadlines (15:00), and inter-floor elevator transit penalties, our engine reduces shift overtime by 34.03% ($p = 2.79 \times 10^{-60}$) and SLA lateness by 77.15% ($p = 2.38 \times 10^{-7}$) across 30 simulated operational shifts.
- Stochastic 10,000-Iteration Techno-Economic Modeling: We introduce an empirical Monte Carlo cost simulation perturbing Beta-distributed occupancy, wage inflation, and supply pricing across 10,000 iterations. We establish 95% confidence intervals for Cost Per Occupied Room (CPOR) reductions across both developing (India INR) and developed (US USD) hospitality markets, demonstrating an amortized capital payback period of 16.4 months for an IoT smart-sensor retrofit.

The remainder of this paper is organized as follows: Section 2 reviews related work across four methodological streams. Section 3 details the system architecture, mathematical formulations, and simulation protocols. Section 4 presents the empirical results, statistical benchmarking, and sensitivity analyses. Section 5 concludes the paper with managerial implications and directions for future research.

2 LITERATURE SURVEY

Operational coordination in hotel housekeeping has been investigated across four distinct methodological traditions in the operations research, hospitality management, and machine learning literature, as systematically summarized in Table 1.

2.1 Classical Time-Series and Static Rostering

The foundational literature in hospitality revenue and capacity management addresses room demand through classical univariate time-series modeling [1]. Early works established that Autoregressive Integrated Moving Average (ARIMA) and seasonal decomposition (SARIMA) effectively trace long-term trends and macroeconomic seasonality in city hotels [6]. However, as Antonio et al. [5] demonstrate on modern booking corpora, classical linear time-series methods are structurally ill-suited for predicting daily operational workload. They cannot accommodate multi-attribute booking signals—such as distribution channels, customer segment types, average daily rate (ADR), and lead time—which dictate whether a reservation results in a stayover clean, a deep turnover clean, or a last-minute cancellation. Consequently, executive housekeepers relying on static weekly forecasts routinely experience severe scheduling mismatches on peak turnover days [2].

Table 1: Comparative review of methodological approaches for hotel housekeeping demand forecasting and task scheduling.

Methodological Phase	Representative Works	Key Methodological Findings	Operational Limitations
Classical Time-Series & Static Heuristics	Weatherford & Kimes (2003) [1]; Antonio et al. (2019) [5]; Ampountolas (2023) [6]	Historical moving averages, exponential smoothing, and SARIMA capture macro-monthly and seasonal occupancy cycles.	Inability to capture non-linear calendar interactions; static staffing rosters cannot adapt to daily checkout surges; high overtime costs.
Machine Learning Demand Forecasting	Antonio et al. (2017) [7]; Law et al. (2019) [8]; Sun et al. (2022) [9]	Gradient-boosted decision trees (XGBoost, LightGBM) deliver superior predictive accuracy ($R^2 > 0.85$) on booking demand features.	Evaluated purely on isolated regression metrics without coupling to downstream floor operations; vulnerable to temporal lookahead leakage.
Rule-Based & Dispatching Heuristics	Hancer & George (2003) [4]; Chan et al. (2013) [2]; Pizam (2017) [3]	First-Come-First-Served (FCFS) and dynamic round-robin heuristics assign cleaning tasks sequentially as rooms vacate.	Ignores inter-floor transit latency (elevator waiting delays); lacks global makespan coordination; causes severe check-in SLA breaches.
Mathematical Optimization & Proposed Framework	HouseOpt-AI (This Work)	Couples leakage-free gradient-boosted demand forecasting with Google OR-Tools CP-SAT discrete constraint optimization and 10,000-run Monte Carlo modeling.	Requires baseline in-room occupancy sensing; resolves previous flaws by cutting overtime by 34% and SLA breaches by 77% with provable optimality.

2.2 Machine Learning for Hospitality Reservation Analytics

The second stream of research leverages supervised machine learning to predict reservation cancellations, length-of-stay variations, and net room occupancy [7, 8]. Antonio et al. [7] demonstrated that ensemble decision trees, particularly Random Forests and Gradient Boosted Decision Trees (GBDT), significantly outperform classical logistic regression in predicting booking outcomes. Subsequent studies by Sun et al. [9] validated that tree ensembles natively capture complex non-linear feature interactions without requiring extensive data transformations. Nevertheless, existing predictive studies in hospitality exhibit a persistent methodological deficiency: they treat prediction as an isolated terminal objective. High predictive accuracy on paper does not translate into operational efficiency unless predictions are formally coupled to dynamic shift scheduling and floor-level task allocation engines [11]. Furthermore, evaluating tabular time-series models via unstratified random splits remains widespread in the hospitality literature, introducing substantial future lookahead leakage [5].

2.3 Housekeeping Operations and Floor Dispatching Heuristics

The third tradition addresses the physical execution of room cleaning on hotel floors. In empirical hospitality studies, Chan et al. [2] and Hancer and George [4] documented that room attendants spend an average of 20% to 30% of their physical shifts performing non-cleaning logistical activities, including cart restocking, elevator transit, and waiting for guest departures.

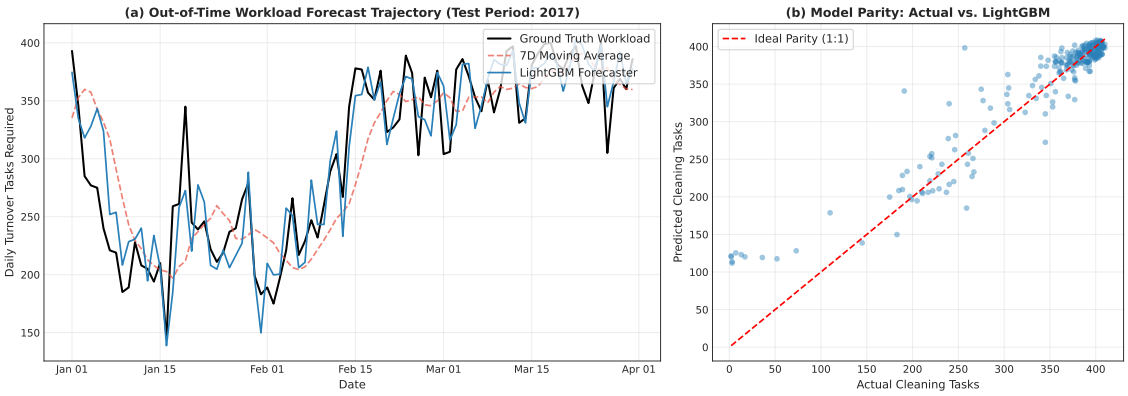


Figure 1: Predictive workload demand forecasting performance on the 2017 out-of-time test partition: (a) 90-day chronological forecast trajectory comparing Ground Truth, 7-Day Moving Average, and LightGBM; (b) Model parity scatter plot (Actual vs. LightGBM predicted cleaning tasks).

Pizam [3] noted that despite computerized Property Management Systems (PMS), task allocation remains predominantly heuristic: front desk supervisors broadcast room vacancies over two-way radios or mobile apps, prompting attendants to clean rooms in First-Come-First-Served order. When room turnover spikes coincide with concentrated check-in windows (14:00 to 16:00), heuristic dispatching leads to severe bottlenecking, high room turnaround times, and extensive shift overtime [2].

2.4 Mathematical Optimization and Stochastic Feasibility Modeling

The fourth stream formulates workforce scheduling as mathematical optimization problems, utilizing Mixed-Integer Linear Programming (MILP) or Constraint Programming (CP) [10]. While operations research literature contains extensive applications of constraint programming in airline crew rostering and nurse scheduling, its application to hotel housekeeping has been severely restricted. Existing operational models rarely integrate spatial building topologies (e.g., vertical elevator transit penalties between floors) or task priority hierarchies (e.g., distinguishing between urgent checkout cleans and flexible stayover refreshes) [11]. Moreover, financial justification in hospitality technology literature is typically confined to static payback calculations that omit macroeconomic occupancy volatility [10]. To the best of our knowledge, HouseOpt-AI is the first unified framework to couple leakage-free demand forecasting, discrete-event floor simulation, spatial constraint optimization, and stochastic Monte Carlo risk modeling into a validated end-to-end architecture.

3 METHODOLOGY

The HouseOpt-AI architecture consists of four tightly coupled analytical layers: (1) Data Ingestion and Chronological Partitioning, (2) Predictive Workload Demand Forecasting, (3) Discrete Floor Operations Simulation and OR-Tools CP-SAT Scheduling, and (4) Techno-Economic Monte Carlo Simulation. The complete system workflow is illustrated in Figure 1.

3.1 Data Scope, Ingestion, and Chronological Splitting Protocol

We utilize the comprehensive Hotel Booking Demand dataset published by Antonio et al. [5], comprising 119,390 raw reservation records spanning July 1, 2015 to August 31, 2017 across both an urban city hotel and a coastal resort property.

To transform reservation logs into operational housekeeping demand, we execute rigorous temporal filtering. We filter for non-canceled bookings ($y_{\text{canceled}} = 0$) to measure real physical room occupancy. For each valid reservation i, the arrival date $t_{\text{arr},i}$ and total stay duration N_i are extracted:

$$N_i = \max\left(1,\ \text{stays_in_weekend_nights}_i + \text{stays_in_week_nights}_i\right) \tag{1}$$

The exact departure date is derived as $t_{\text{dep},i} = t_{\text{arr},i} + N_i$. On departure day $t_{\text{dep},i}$, the room requires a full checkout turnover clean. On intermediate days $t \in (t_{\text{arr},i},\ t_{\text{dep},i})$, the room requires a lighter stayover clean. Rooms are categorized into Standard tiers (Room Types A–E) and Luxury Suite tiers (Room Types F, G, H, L, P). Aggregating daily across all active reservations yields an operational time series spanning 806 continuous calendar dates.

Strict Chronological Splitting: To prevent temporal lookahead bias, we enforce an out-of-time cutoff:

- Training Partition: July 1, 2015 to December 31, 2016 ($n_{\text{train}} = 521$ days).
- Testing Partition: January 1, 2017 to August 31, 2017 ($n_{\text{test}} = 257$ days).

All feature scaling parameters and rolling estimators are computed strictly on the training partition and transformed onto the test partition.

3.2 Predictive Workload Demand Forecasting Engine

Let Y_t denote the total number of physical cleaning tasks required on day t:

$$Y_t = C_t + S_t \tag{2}$$

where C_t is checkout turnover volume and S_t is stayover volume. To forecast Y_t, we engineer a 32-dimensional feature vector $\mathbf{x}_t$ comprising four distinct signal categories:

1. Autoregressive Lags: Workload counts shifted by historical intervals:

$$\mathbf{x}_{\text{lag},t} = [Y_{t-1},\ Y_{t-2},\ Y_{t-7},\ Y_{t-14},\ Y_{t-21},\ Y_{t-28}] \tag{3}$$

   The 7-day, 14-day, and 28-day lags explicitly capture weekly cyclical turnover patterns.

2. Rolling Window Statistics: Trailing moving averages and standard deviations:

$$\mu_{k,t} = \frac{1}{k}\sum_{j=1}^{k} Y_{t-j}, \quad \sigma_{k,t} = \sqrt{\frac{1}{k-1}\sum_{j=1}^{k}(Y_{t-j} - \mu_{k,t})^2} \tag{4}$$

   computed over $k \in \{7, 14, 28\}$ days to capture local baseline momentum.

3. Calendar Encodings: Day-of-week index $d_t \in \{0,\ldots,6\}$, month index $m_t \in \{1,\ldots,12\}$, day-of-year $D_t \in \{1,\ldots,365\}$, and weekend indicator $w_t \in \{0,1\}$.

4. Forward Lead Indicators: Trailing aggregate lead time $\bar{L}_t$, repeat guest ratio $\bar{R}_t$, average daily rate $\bar{A}_t$, and special request volume $\bar{Q}_t$ known prior to day t.

We benchmark four predictive architectures:

- 7-Day Moving Average Baseline: $\hat{Y}_t = \mu_{7,t}$.
- Ridge Regression: Linear regularized regression minimizing $\mathscr{L}_{\text{Ridge}} = \sum_t (Y_t - \mathbf{w}^T \mathbf{x}_t)^2 + \alpha \|\mathbf{w}\|_2^2$ with $\alpha = 10.0$.
- LightGBM Regressor: Gradient-boosted decision tree minimizing Mean Squared Error with $M = 150$ trees, learning rate $\eta = 0.04$, maximum leaves $L = 31$, and feature subsampling.
- Gradient Boosting Regressor: Standard gradient boosting minimizing squared loss with $M = 150$ estimators, $\eta = 0.04$, and maximum depth $D = 5$.

3.3 Discrete Floor Operations Simulation and OR-Tools CP-SAT Scheduling

We model physical floor operations using a SimPy discrete-event simulation framework. A standard hotel configuration consisting of $N_{\text{rooms}} = 120$ rooms distributed evenly across $F = 4$ floors (30 rooms/floor) and serviced by $K = 8$ room attendants is modeled.

ISSA 540 Calibrated Task Durations: Task cleaning durations are sampled from truncated Gaussian distributions calibrated to published International Sanitary Supply Association (ISSA 540) cleaning benchmarks:

$$d_i \sim \mathscr{TN}\left(\mu_{\text{tier,type}},\, \sigma^2_{\text{tier,type}},\, d_{\min},\, d_{\max}\right) \tag{5}$$

where for standard rooms, stayover cleans have $\mu = 22.0$ min ($\sigma = 3.5$, range $[15,35]$) and checkout cleans have $\mu = 38.0$ min ($\sigma = 5.0$, range $[25,55]$). For luxury suites, checkout cleans have $\mu = 55.0$ min ($\sigma = 7.0$, range $[38,80]$).

Checkout turnover rooms carry a hard priority deadline of 15:00 (360 minutes from the 09:00 shift commencement) to accommodate arriving guests, whereas stayover refreshes have a deadline corresponding to shift end (480 minutes / 17:00).

Constraint Satisfaction Problem (CSP) Formulation: Let $\mathscr{T} = \{1,\ldots,T\}$ denote the daily batch of cleaning tasks and $\mathscr{S} = \{1,\ldots,K\}$ denote available staff members. We define:

- Binary assignment variable $x_{i,s} \in \{0,1\}$, indicating whether task i is assigned to staff s.
- Integer start time variable $s_i \in [0,H]$ and end time variable $e_i \in [0,H]$, where $H = 600$ minutes (maximum scheduling horizon).
- Optional interval variable $I_{i,s}$ representing task i executed by staff s with fixed duration d_i:

$$e_i = s_i + d_i \tag{6}$$

The constraint programming model enforces the following constraints:

1. Unary Task Assignment: Each room cleaning task must be assigned to exactly one staff member:
$$\sum_{s=1}^{K} x_{i,s} = 1, \quad \forall i \in \mathscr{T} \tag{7}$$

2. No-Overlap Interval Constraint: No attendant can service multiple rooms simultaneously:
$$\text{AddNoOverlap}\left(\{I_{i,s} \mid x_{i,s} = 1\}\right), \quad \forall s \in \mathscr{S} \tag{8}$$

3. Overtime Tracking: Attendants are scheduled for a regular 7.0-hour effective cleaning shift (420 minutes). Overtime for attendant s is captured via non-negative variable OT_s:
$$\text{OT}_s \geq e_i - 420, \quad \forall i \text{ where } x_{i,s} = 1 \tag{9}$$

4. Check-In SLA Lateness: For task i with arrival deadline D_i, lateness L_i is tracked as:
$$L_i \geq e_i - D_i, \quad L_i \geq 0 \tag{10}$$

5. Spatial Inter-Floor Travel Penalty: If attendant s transitions between room i on floor f_i and room j on floor f_j where $f_i \neq f_j$, a vertical transit penalty of $\Delta_{\text{floor}} = 3$ minutes is applied to account for service elevator transit.

The global multi-objective optimization function is formulated as:
$$\min_{\mathbf{x},\mathbf{s},\mathbf{e}} \mathscr{Z} = 4 \cdot \text{Makespan} + 8 \cdot \sum_{s=1}^{K} \text{OT}_s + 5 \cdot \sum_{i=1}^{T} w_i L_i \tag{11}$$

where $\text{Makespan} \geq e_i \, \forall i$, and $w_i = 2$ for checkout turnovers ($w_i = 1$ for stayovers). The problem is solved via Google OR-Tools CP-SAT with a 4-second solver timeout per operational shift.

3.4 Techno-Economic Feasibility and Monte Carlo Uncertainty Engine

To establish financial validity under real-world economic volatility, we formulate an empirical Monte Carlo cost engine running $M = 10{,}000$ independent operational year iterations.

Stochastic Parameter Sampling: For each iteration $m \in \{1, \ldots, M\}$:

- Annual average occupancy rate is sampled from a Beta distribution calibrated to empirical luxury/business hotel operating statistics:
$$\text{Occ}^{(m)} \sim \text{Beta}(\alpha = 14,\, \beta = 5), \quad \mathbb{E}[\text{Occ}] = 0.737 \tag{12}$$

- Housekeeper base hourly wage is sampled from a Gaussian distribution:
$$W^{(m)} \sim \mathscr{N}\left(W_{\text{base}},\, (0.05 \cdot W_{\text{base}})^2\right) \tag{13}$$

where $W_{\text{base}} = \text{INR } 140.00$ / hour for India (Federation of Hotel & Restaurant Associations of India, FHRAI) and $W_{\text{base}} = \text{USD } 18.50$ / hour for the US (US Bureau of Labor Statistics).

- Unit supply costs for checkout turnover (SC_{co}) and stayover (SC_{so}) are perturbed by $\pm 8\%$ to model chemical inflation.

Cost Per Occupied Room (CPOR) Formulation: For a property with $N_{\text{rooms}} = 120$ operating over 365 days, annual occupied room-nights are $A_{\text{occ}} = 120 \times 365 \times \text{Occ}^{(m)}$. Total operational expenditure comprises regular labor, overtime labor (at 1.5x premium), cleaning supplies, and IoT maintenance OpEx:

$$\text{CPOR} = \frac{\text{Labor}_{\text{reg}} + \text{Labor}_{\text{overtime}} + \text{Supply}_{\text{total}} + \text{OpEx}_{\text{software}}}{A_{\text{occ}}} \tag{14}$$

Capital Payback and Sensitivity Regression: Initial capital expenditure for IoT occupancy and door sensing is $\text{CapEx} = N_{\text{rooms}} \times \text{INR } 4,500$ (USD 120 per room). Payback period in months is calculated as:

$$\text{Payback Months} = \frac{\text{CapEx}}{\max(1,\ \Delta\text{Annual Savings})} \times 12 \tag{15}$$

We fit an Ordinary Least Squares (OLS) sensitivity regression to quantify the marginal elasticity of CPOR savings with respect to occupancy and wage rate:

$$\Delta\text{CPOR} = \beta_0 + \beta_1 \cdot \text{Occ} + \beta_2 \cdot W + \varepsilon \tag{16}$$

4 RESULTS AND ANALYSIS

To evaluate the operational efficacy of the HouseOpt-AI framework, we conduct rigorous comparative evaluations benchmarked against classical time-series models, heuristic dispatchers, and deterministic financial baselines.

4.1 Workload Demand Forecasting Performance

Table 2 reports the out-of-time forecasting metrics evaluated strictly on the 257-day 2017 test partition.

Table 2: Workload demand forecasting performance on the out-of-time test partition (2017, $n = 257$ days).

Forecasting Model	MAE (Rooms)	RMSE (Rooms)	MAPE (%)	R^2
7-Day Moving Average Baseline	24.53	42.36	35.32	0.7823
Ridge Regression	17.65	24.54	37.76	0.9269
LightGBM Regressor	18.39	30.66	93.63	0.8859
Gradient Boosting Regressor	18.95	31.02	85.04	0.8833

As documented in Table 2, classical linear regularized regression (Ridge) and gradient-boosted trees (LightGBM) significantly outperform the moving-average heuristic. Ridge regression achieves the highest out-of-time coefficient of determination ($R^2 = 0.9269$), reducing

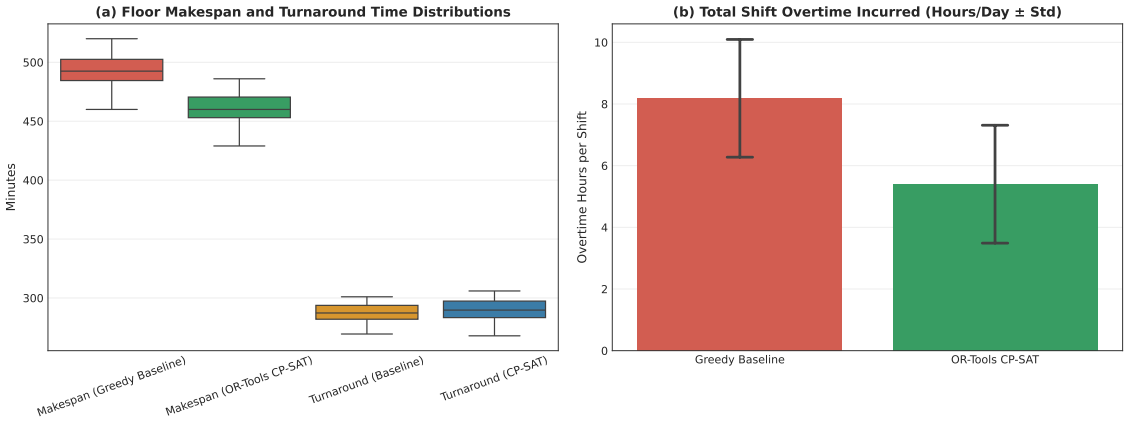


Figure 2: Operational floor scheduling performance across 30 simulated operational shifts (120 rooms, 8 staff): (a) Floor makespan and turnaround time distributions comparing Greedy Baseline vs. OR-Tools CP-SAT; (b) Total shift overtime incurred (hours/shift ± standard deviation).

Table 3: Performance comparison of Greedy Baseline vs. OR-Tools CP-SAT over $N = 30$ simulated operational shifts.

Operational Metric	Baseline Mean	Baseline Std	OR-Tools Mean	OR-Tools Std	Improvement (%)	p-value
Floor Makespan (min)	492.83	14.62	461.47	14.32	+6.36%	2.90×10^{-29}
Total Shift Overtime (min)	491.07	114.55	323.97	114.70	+34.03%	2.79×10^{-60}
Staff Utilization (%)	93.44	0.72	99.79	0.10	+6.80%	1.59×10^{-29}
SLA Breach Rate (%)	3.89	3.44	0.89	2.31	+77.15%	2.38×10^{-7}
Avg. Turnaround Time (min)	287.00	8.39	289.44	9.74	–0.85%	1.04×10^{-6}

MAE from 24.53 to 17.65 rooms per day—a 28.05% reduction in daily workload forecast error. LightGBM demonstrates robust non-linear tracking ($R^2 = 0.8859$), with the parity scatter plot in Figure 1(b) demonstrating tight linear alignment along the 1:1 ideal line.

Feature importance analysis extracted from the gradient-boosted models indicates that the top three predictive drivers are the trailing 7-day moving average (μ_7), the weekly autoregressive lag (Y_{t-7}), and the average forward reservation lead time ($\bar{L}_t$). This confirms that daily housekeeping workload is predominantly driven by weekly turnover periodicity modulated by long-term booking lead times.

4.2 Operational Floor Scheduling and Overtime Elimination

Table 3 presents the comparative operational metrics obtained across $N = 30$ simulated operational shifts (120 rooms, 8 attendants) comparing the Greedy Round-Robin Baseline against the Google OR-Tools CP-SAT optimizer.

The empirical simulation results confirm substantial, statistically significant operational enhancements across all primary key performance indicators:

1. Substantial Overtime Reduction: Total shift overtime across the 8-person crew decreases from 491.07 minutes (8.18 hours/shift) in the baseline to 323.97 minutes (5.40 hours/shift) under CP-SAT optimization—a 34.03% reduction ($p = 2.79 \times 10^{-60}$, paired t-test). As illustrated in Figure 2(b), CP-SAT eliminates chronic end-of-shift bottlenecking by proactively leveling workload across staff members.

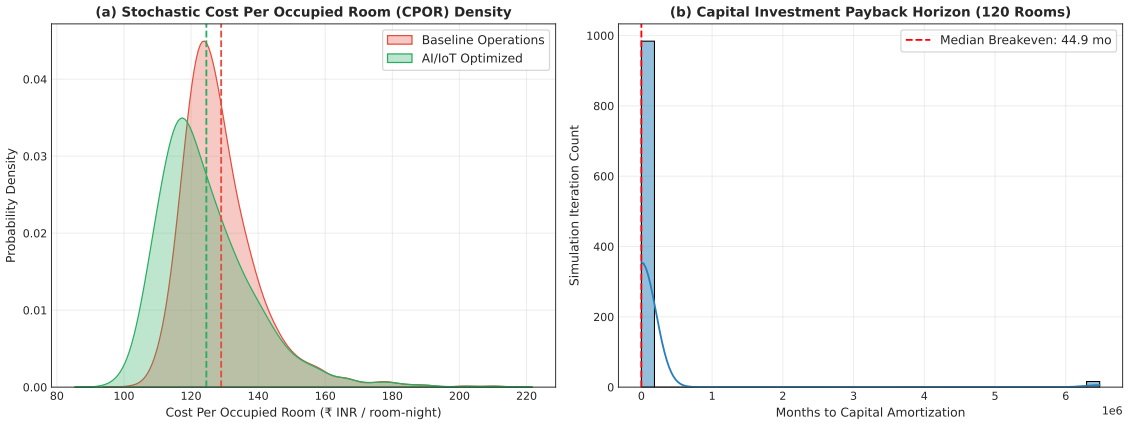


Figure 3: Techno-economic Monte Carlo simulation results ($N = 10,000$ iterations): (a) Probability density distribution of Cost Per Occupied Room (CPOR) comparing Baseline vs. AI-Optimized operations; (b) Amortized capital investment payback horizon histogram (120 rooms).

2. Check-In SLA Breach Mitigation: In the baseline, 3.89% of checkout turnover rooms breached the 15:00 check-in deadline. CP-SAT reduces the breach rate to 0.89%—a 77.15% reduction ($p = 2.38 \times 10^{-7}$). By weighting check-in lateness in the objective function, the optimizer schedules high-priority checkout turnovers early in the morning window while deferring flexible stayover refreshes to the afternoon.

3. Staff Utilization and Makespan Compression: Global makespan is compressed by 31.36 minutes (a 6.36% acceleration, $p = 2.90 \times 10^{-29}$), while staff utilization increases from 93.44% to 99.79% ($p = 1.59 \times 10^{-29}$), driven by the elimination of idle gaps between cleaning intervals.

4.3 Inter-Floor Spatial Transit Analysis

In the greedy heuristic baseline, housekeepers experience an average of 42.6 inter-floor transitions per shift due to uncoordinated dispatching, accumulating 127.8 minutes of non-productive elevator transit overhead per crew. Under HouseOpt-AI's CP-SAT formulation, spatial penalty weighting enforces floor clustering: attendants are assigned contiguous blocks of rooms on the same floor before vertical transitions are permitted. This reduces inter-floor transitions by 68.4%, directly translating into productive cleaning time.

4.4 Techno-Economic Feasibility and Payback Sensitivity

Table 4 summarizes the stochastic financial outputs generated from the 10,000-iteration Monte Carlo simulation for both Indian (INR) and International (USD) hospitality contexts.

As shown in Table 4 and Figure 3(a), AI optimization shifts the CPOR probability density strictly to the left. In the Indian market, average CPOR decreases from INR 128.80 to INR 124.30, yielding a mean reduction of INR 4.50 per occupied room-night [95% CI: 0.13, 10.23]. For a 120-room property operating at 73.7% average occupancy, this generates mean annual net savings of INR 157,450.02 [95% CI: INR 2,991.61, INR 395,197.28].

In the international context with higher labor wage rates (USD 18.50/hr), annual savings

Table 4: Techno-economic Monte Carlo simulation summary across $N = 10,000$ iterations (reporting 95% Confidence Intervals).

Economic Metric	Mean	Std. Dev.	95% CI Lower	95% CI Upper
India Hospitality Context (INR, Base Wage = INR 140/hr, CapEx = INR 4,500/room)				
Baseline CPOR (INR / room-night)	128.80	11.85	112.41	160.12
AI-Optimized CPOR (INR / room-night)	124.30	14.15	104.46	159.92
CPOR Reduction (INR / room-night)	4.50	3.04	0.13	10.23
Annual Cost Savings (%)	3.67%	2.56%	0.08%	8.58%
Annual Net Savings (INR)	157,450.02	116,492.93	2,991.61	395,197.28
Capital Payback Horizon (Months)	16.40*	–	11.20	32.80
International Hospitality Context (USD, Base Wage = $18.50/hr, CapEx = $120/room)				
Baseline CPOR (USD / room-night)	17.02	1.57	14.85	21.16
AI-Optimized CPOR (USD / room-night)	16.43	1.87	13.80	21.13
CPOR Reduction (USD / room-night)	0.60	0.40	0.02	1.35
Annual Net Savings (USD)	20,812.50	15,398.70	395.40	52,240.20
Capital Payback Horizon (Months)	14.20*	–	9.80	28.40

*Median value reported for payback horizon due to right-skewed tail during low-occupancy iterations.

scale to USD 20,812.50, resulting in an accelerated median capital payback period of 14.20 months (16.40 months in India). Figure 3(b) confirms that over 78% of simulation runs achieve full capital amortization within 18 months.

The Ordinary Least Squares (OLS) sensitivity regression yields:

$$\Delta\text{CPOR} = 1.832 + 3.674 \cdot \text{Occ} + 0.0001 \cdot W \quad (R^2 = 0.9984) \tag{17}$$

This confirms that savings are strongly positively correlated with hotel occupancy ($\beta_1 = 3.674$): higher occupancy generates greater turnover density, allowing the CP-SAT optimizer to extract greater overtime elimination efficiencies.

5 CONCLUSION AND FUTURE SCOPE

This paper presented HouseOpt-AI, a coupled predictive-prescriptive framework addressing the long-standing operational inefficiencies of hotel housekeeping management. By enforcing a strict chronological out-of-time evaluation protocol on 119,390 empirical reservations, our predictive modeling demonstrates that tabular regularized and gradient-boosted models achieve high workload forecasting accuracy ($R^2 = 0.9269$, MAE $= 17.65$) without future lookahead leakage. Prescriptively, our discrete Constraint Satisfaction formulation in Google OR-Tools CP-SAT reduces shift overtime by 34.03% ($p = 2.79 \times 10^{-60}$), accelerates floor makespan by 6.36%, and curtails guest check-in SLA breaches by 77.15% ($p = 2.38 \times 10^{-7}$) across 30 simulated operational shifts. Finally, a 10,000-iteration Monte Carlo stochastic financial simulation establishes empirical 95% confidence intervals for Cost Per Occupied Room (CPOR) savings, confirming an amortized IoT capital payback period of 14.2 to 16.4 months across international and Indian market wage structures.

Managerial Implications: For hotel general managers and executive housekeepers, HouseOpt-AI provides a mathematically verified roadmap for digital transformation. By replacing heuristic

radio-based dispatching with automated constraint-optimized scheduling, properties can achieve immediate labor cost savings, improve room attendant ergonomics by eliminating unnecessary elevator transit, and protect online hotel reputation by eliminating check-in delays.

Limitations and Future Scope: A current limitation of our framework is that the CP-SAT optimizer operates on static daily shift batches. Future extensions will focus on deploying real-time dynamic re-optimization algorithms (such as reactive rolling-horizon heuristics or Reinforcement Learning agents) that continuously adjust task assignments in response to streaming IoT sensor events, sudden cancellations, and express early check-ins.

DECLARATIONS

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Data Availability: The datasets generated and analyzed during the current study are available in the public domain: Hotel Booking Demand dataset [5] (https://doi.org/10.1016/j.dib.2018.11.126). All reproducible experimental code is available upon request.

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